

BAYES' THEOREM :

- Reverend Thomas Bayes.
- It deals with "updating the probability of an event based on new evidence / information."

$$P(A|B) = \frac{P(B|A) \times P(A)}{P(B)}$$

- * This formula helps us update the probability of an event (A) based on new evidence B.

where, $P(A|B)$ = updated probability of an event after taking new evidence into account.

$P(A) = P_{prior}$: Initial belief about A before seeing B.

$P(A)$ = Marginal : Initial belief about A before seeing B.
 $P(B)$ = Marginal : Overall probability of B (used to normalize the result).

$P(B|A)$ = likelihood : Evidence that supports the initial belief.

* USECASES :-

- Medical Diagnosis
- Spam Classification
- Recommendation Systems
- Fraud Detection

#	CONCEPT :
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$\Rightarrow$ From CONDITIONAL PROBABILITY we know that:-

$$P(A|B) = \frac{P(A \cap B)}{P(B)}, \text{ also } \dots\dots\dots (1)$$

$$P(B|A) = \frac{P(B \cap A)}{P(A)} \quad \text{--- (2)}$$

∴ From eqn (i) & (ii) we have :-

$$P(A|B) \cdot P(B) = P(A \cap B) \quad \text{En} \quad \dots\dots\dots (3)$$

$$P(B|A) \cdot P(A) = P(B \cap A) \quad \text{--- (4)}$$

∴ RHS is equal so we have :-

$$P(A \cap B) = P(A|B) \cdot P(B) = P(B|A) \cdot P(A)$$

$$\Rightarrow P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$$

Posterior

Likelihood

Marginal

Prior

* Example :- From a deck of 52 cards, find the probability that the drawn face cards is a king card.

Soln :- Alg - $P(\text{king} | \text{Face}) = ?$

∴ Using Bayes' Theorem we know :-

$$P(\text{king} | \text{Face}) = \frac{P(\text{Face} | \text{king}) \cdot P(\text{king})}{P(\text{Face})}$$

$$= \frac{(1 \times 4/52)}{(12/52)}$$

$$= \frac{(1 \cdot 1/13)}{(3/13)}$$

$$= \frac{1}{3}$$